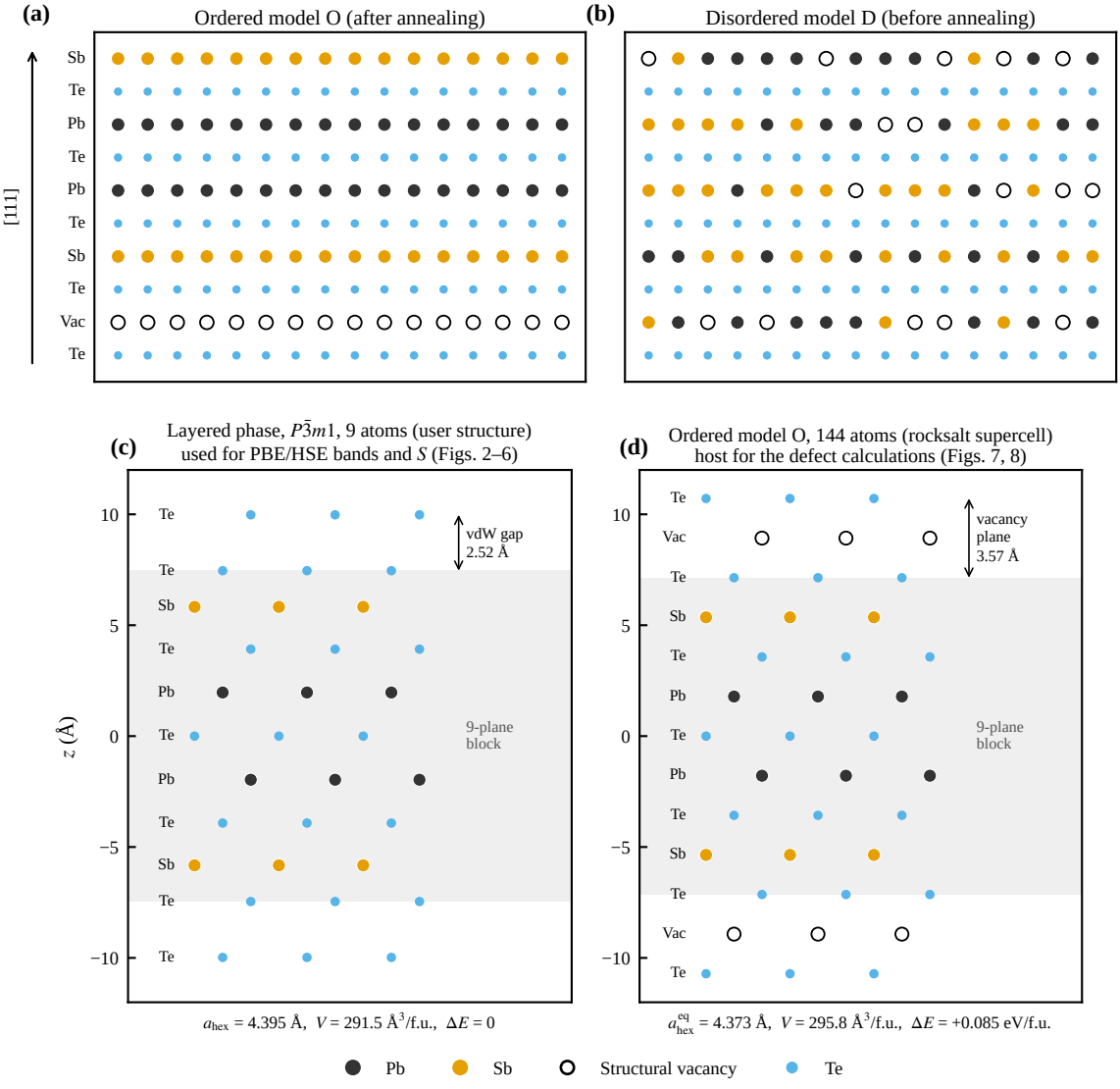

(a)
Ordered model O (after annealing)
(b)
Disordered model D (before annealing)
[111]
Sb
Te
Pb
Te
Pb
Te
Sb
Te
Vac
Te
(c)
Layered phase, $P\bar{3}m1$, 9 atoms (user structure)
used for PBE/HSE bands and S (Figs. 2–6)
vdW gap
2.52 Å
9-plane
block
z (Å)
10
5
0
−5
−10
a_{hex} = 4.395 Å, V = 291.5 Å^3/f.u., ΔE = 0
(d)
Ordered model O, 144 atoms (rocksalt supercell)
host for the defect calculations (Figs. 7, 8)
vacancy
plane
3.57 Å
9-plane
block
a_{hex}^{eq} = 4.373 Å, V = 295.8 Å^3/f.u., ΔE = +0.085 eV/f.u.
Pb
Sb
Structural vacancy
Te